

RETAIL SALES & CUSTOMER ANALYTICS

A MySQL + Python + Power BI Project

Prepared by Mohan · Data Analyst Portfolio Project

Business Question

Superstore's leadership wants to know: **where is the business losing money, which customers are at risk of leaving, and why?**

Specifically, this project set out to answer three things:

1. Is revenue growing, and how volatile is it month to month?
2. Which customers are most valuable — and which are at risk of churning?
3. Which product categories are least profitable, and why?

Approach

1. Database Design (MySQL)

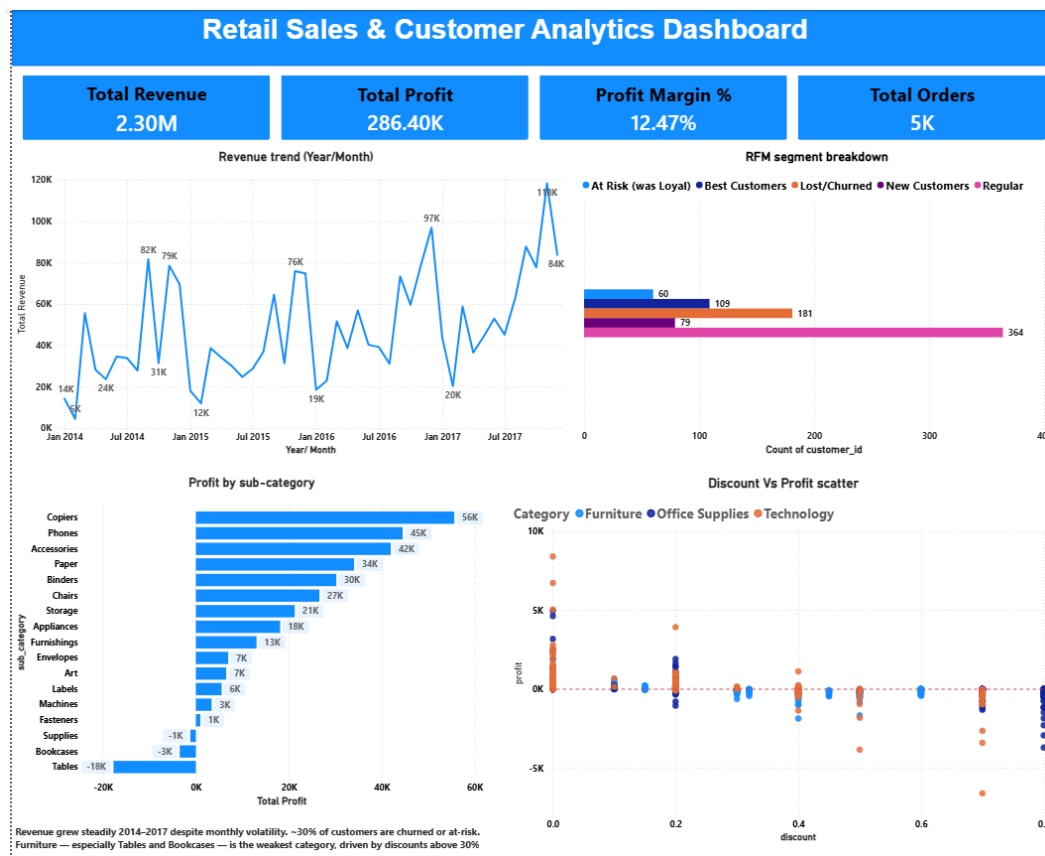
Rather than working off a single flat CSV, a normalized relational schema was designed with four related tables — customers, products, orders, and order_details — connected via primary and foreign keys. Raw data was staged, cleaned (handling mixed date formats and encoding issues), and loaded into the normalized structure using a mix of SQL and a Python-based ETL script, built after working through Windows-specific MySQL import and driver issues.

2. Analysis (SQL + Python)

- Wrote SQL queries using joins, GROUP BY, and window functions (NTILE) to calculate monthly revenue trends, top products by profit, and category/region profit margins
- Built an RFM (Recency, Frequency, Monetary) customer segmentation model using CTEs and window functions, classifying all 793 customers into five segments: Best Customers, New Customers, At Risk, Lost/Churned, and Regular
- In Python, pulled this data via SQLAlchemy into Pandas to run a discount-vs-profit correlation analysis and build a cohort retention heatmap tracking repeat purchase behavior over 47 months

3. Dashboard (Power BI)

Connected Power BI directly to the MySQL database via ODBC, built a relational data model matching the MySQL schema, wrote DAX measures (Total Revenue, Profit Margin %, Total Orders), and built an interactive, cross-filterable dashboard combining all findings into one page.



Key Metrics

\$2.30M Total Revenue	793 Customers Segmented	~30% Churned / At-Risk
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Key Findings

Revenue is growing but volatile. Total revenue reached \$2.30M across the dataset, growing steadily year-over-year (2014 → 2017), but with sharp month-to-month swings — e.g., a drop from \$82K to \$12K within a few months — indicating seasonality or inconsistent demand rather than steady growth.

~30% of the customer base is churned or at risk. RFM segmentation revealed 181 customers (23%) as fully churned and 60 (8%) as "at risk" — previously loyal customers who have gone quiet. Combined, that's nearly a third of the customer base representing a retention opportunity.

Furniture is the weakest category, driven by discounting. Furniture showed negative profit margins in 2 of 4 regions (Central: -67%, East: -9%), with average discounts of 17.4% — the highest of any category — compared to just 13.2% for Technology, which remained the most profitable category despite lower average discounts. A discount-vs-profit correlation of -0.22 confirmed this relationship holds across the dataset, though it's notably stronger within Furniture specifically.

Recommendations

1. Cap Furniture discounts below the ~30% threshold where profit consistently turns negative, particularly for Tables and Bookcases (the two least profitable sub-categories, at -\$18K and -\$3K respectively)
2. Launch a win-back campaign targeted at the 241 customers in the At-Risk and Lost/Churned segments, prioritizing At-Risk customers first since they have a proven purchase history
3. Investigate revenue volatility further — determine whether the sharp monthly swings are seasonal (predictable) or indicate inconsistent demand generation that could be smoothed with better marketing cadence

Tools & Skills Demonstrated

MySQL (schema design, joins, CTEs, window functions) · **Python** (Pandas, SQLAlchemy, Seaborn, correlation & cohort analysis) · **Power BI** (DAX, data modeling, ODBC connectivity, interactive dashboards)